Group 01

IT INFRASTRUCTURE AND EMERGING TECHNOLOGIES

GROUP - 01

MANAGEMENT INFORMATION SYSTEM

CST 344- 2

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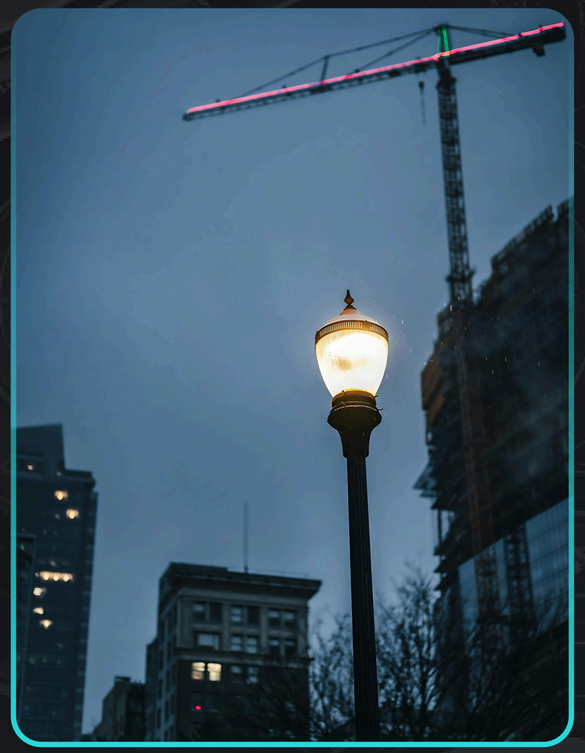
What is

IT Infrastructure & Emerging technology

The basic systems and structures that support a society or organization, enabling it to function efficiently



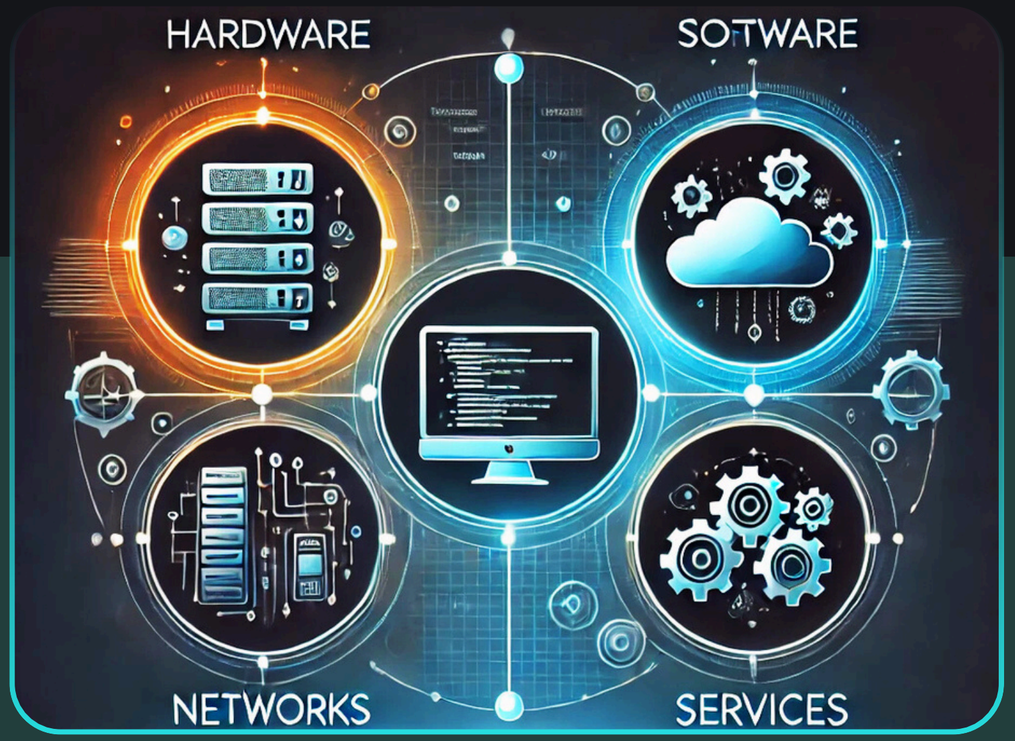




The shared technology platform that enables business operations

Including :

- Hardware
- Software
- Networks
- Services

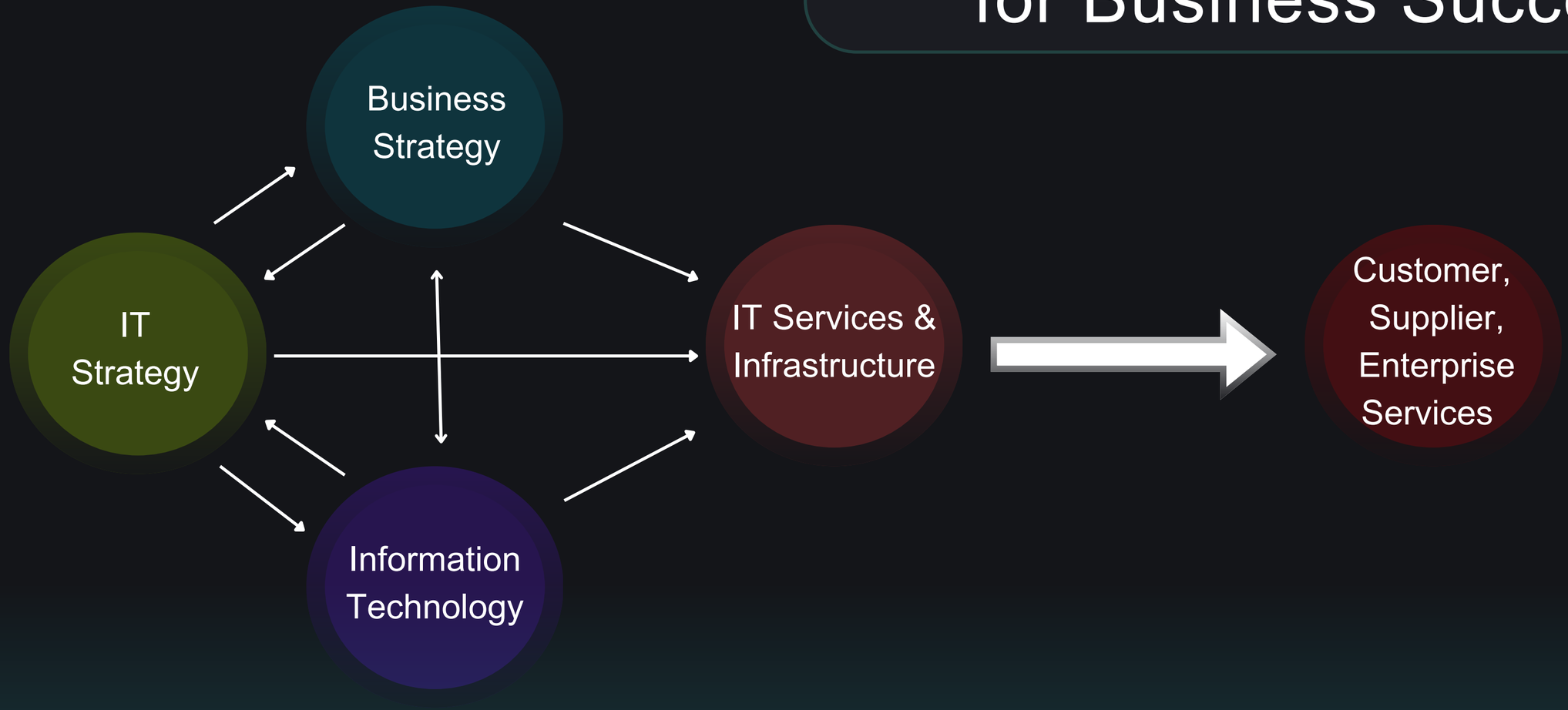


Includes innovations like AI, IoT, and cloud computing that enhance IT infrastructure and drive digital transformation





Aligning IT Infrastructure for Business Success



Importance of IT Standards

What are IT Standards?

- Agreed rules and guidelines for IT systems.
- Ensure compatibility and reliability

Why Are They Important?

- Interoperability: Systems work together.
- Efficiency: Reduces complexity.
- Cost Savings: Simplifies integration.
- Scalability: Supports future growth.

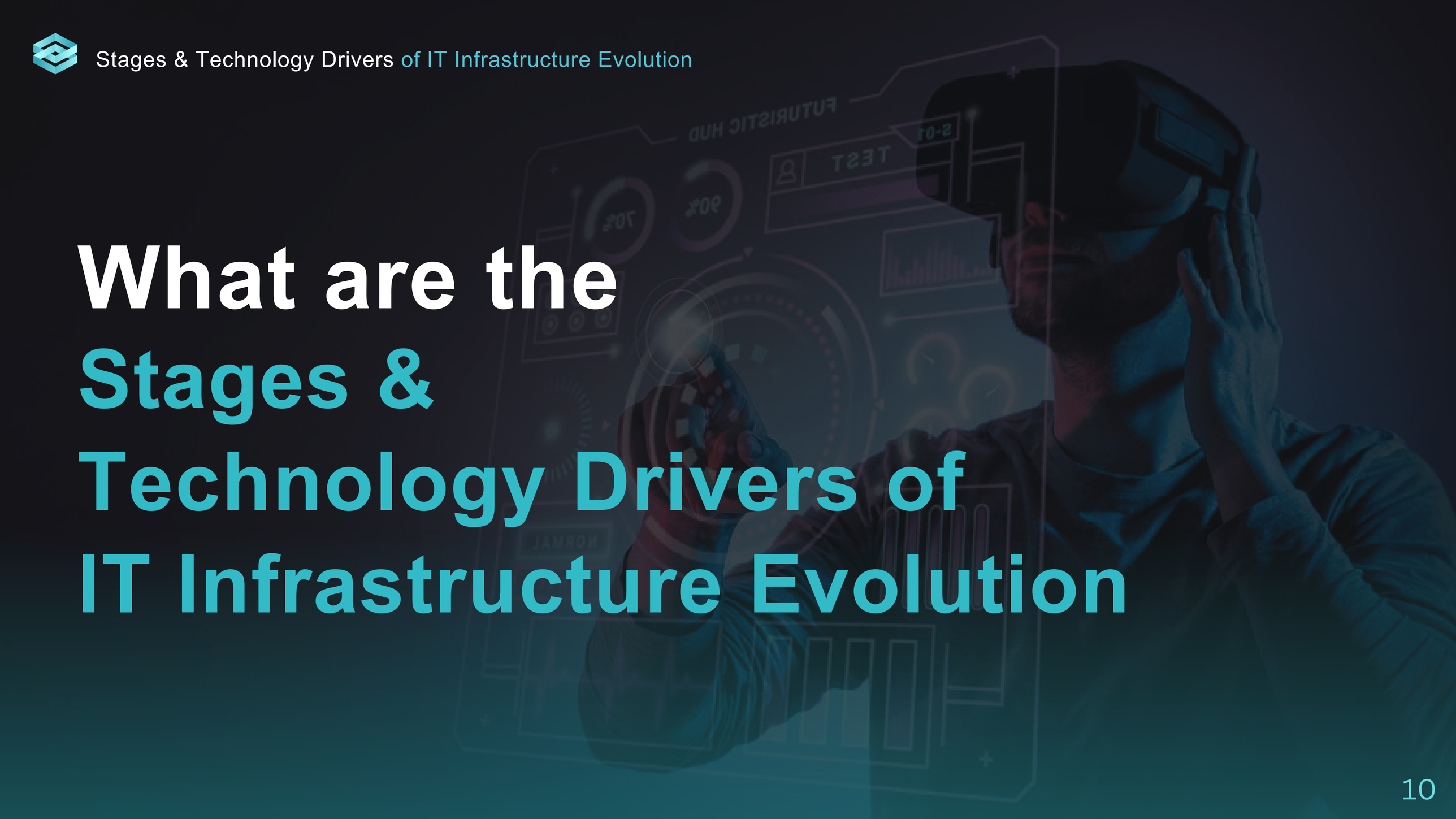
Importance of IT Standards

Examples of IT Standards

- Hardware: USB, Wi-Fi
- Software: HTML, HTTP
- Data: JSON, XML

Benefits to IT Infrastructure

- Enhances security.
- Simplifies management.
- Enables global collaboration.



What are the Stages & Technology Drivers of IT Infrastructure Evolution

What is Evolution of IT Infrastructure?





The evolution of IT infrastructure is driven by several key technological advancements that enable faster more efficient and interconnected systems

Stages of Evolution

1. General-Purpose Mainframe and Minicomputer Era
2. Personal Computers Era
3. Client/Server Networks Era
4. Enterprise Computing Era
5. Cloud and Mobile Computing Era

General Purpose Mainframe & Mini Computer Era



The Minicomputer Era (1960s-1980s) was a period when smaller, affordable computers became popular, offering decentralized computing for medium-sized businesses and research. These systems supported multiple users and tasks, bridging the gap between mainframes and personal computers.

Benefits

- High Processing Power:

Mainframes handled complex calculations and large-scale data processing efficiently.

- Centralized Control:

Simplified data management and resource allocation for large organizations.

- Reliability:

Designed for uninterrupted operation, making them ideal for critical tasks like banking and government operations.

Challenges

- High Cost:

Extremely expensive to purchase, operate, and maintain, limiting access to large organizations.

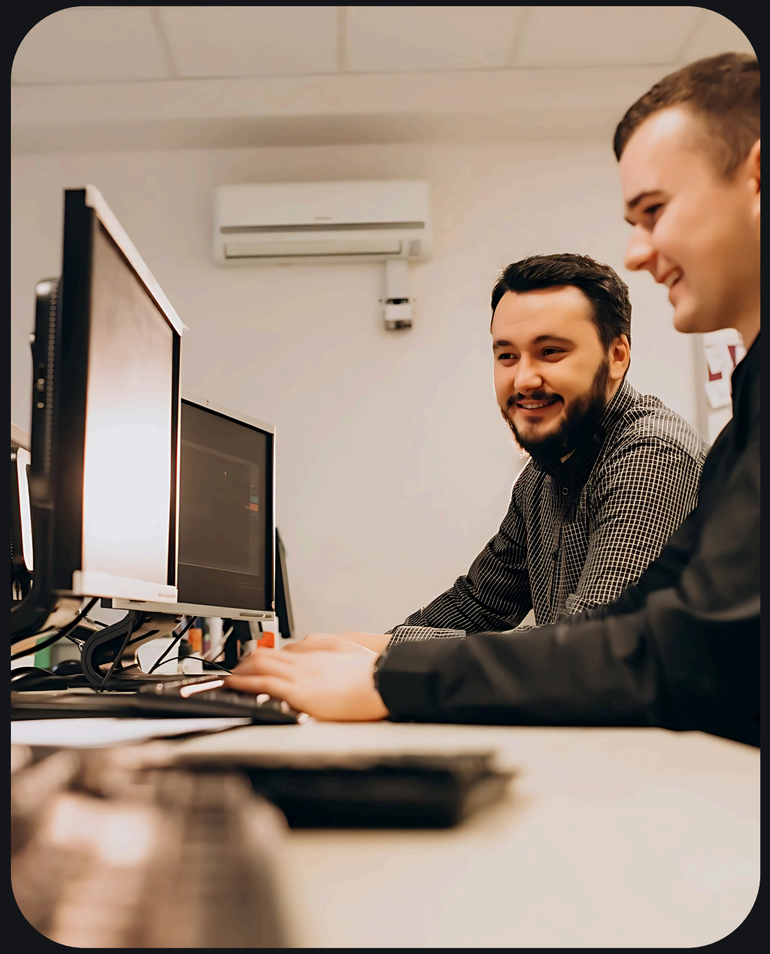
- Complex Operation:

Required specialized personnel for programming, maintenance, and operation.

- Size and Space:

Occupied large physical spaces, needing special facilities for installation and cooling.

Personal Computer Era



- PCs brought computing to individuals and small businesses.
- Decentralized computing became widespread as individuals could own computers
- Software became more user-friendly, with operating systems like MS-DOS and later Windows.
- The PC era transformed workplaces, education, and personal lives, enabling tasks like word processing, spreadsheets, and internet access.
- Networks like LAN (Local Area Networks) enabled PC communication.

Benefits

- Portability:

With advancements in hardware, laptops and other portable devices became common.

- Ease of Use:

User-friendly interfaces (e.g., Windows, MacOS) made technology accessible to non-experts.

- Wide Application:

PCs supported diverse uses, including office work, gaming, education, and multimedia.

Challenges

- Security Risks:

Increased adoption led to challenges like malware, hacking, and data theft.

- Cost of Maintenance:

While affordable to buy, maintaining and upgrading hardware and software added costs.

- Digital Divide:

Not everyone could afford or access PCs, leading to gaps in technological equity.

Client/Server Networks Era (1983-Present)

In this Era where desktop or laptop computers are connected to powerful server computers.Clients act as user entry points, while servers handle data processing, storage and management tasks.

Key Characteristics

- Client-Server Architecture: Two-tier (simple) or N-tier (complex).
- Split Processing: Clients handle user interface, servers manage data and resources.
- Modern Servers: Powerful PCs with multiple processors.

Benefits

- Cost-effective and scalable systems.
- Distributed workloads for better performance.
- Supports multitiered e-commerce and business apps.

Challenges

- Requires robust networking.
- Potential latency in multitier models.



Enterprise Computing Era (1992-Present)

An era of integrating all the systems in a company into one big, connected network to make operations more efficient and seamless.

Key Characteristics

- Enterprise Integration: Unified systems for HR, finance, and inventory.
- ERP and CRM Tools: Centralized data and enhanced customer management.
- Technology Growth: Rise of client-server computing and networked systems.

Benefits

- Improved efficiency through connected systems.
- Cost savings by reducing duplicate systems.
- Faster decision-making with shared data access.

Challenges

- Complex management of integrated systems.
- High implementation and maintenance costs.
- Security risks and the need for constant updates.

Cloud & Mobile Computing Era (2000 - Present)

Cloud computing provides shared resources like storage and software over the internet.

Types of Cloud Services

- IaaS: Rent virtual hardware (Ex: AWS).
- PaaS: Platforms for developers (Ex: Google App Engine).
- SaaS: Software delivered online (Ex: Google Workspace).

Mobile Platforms

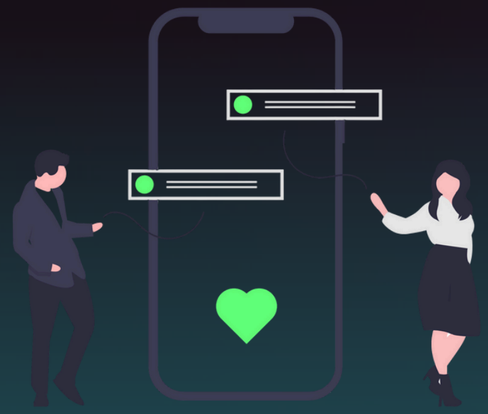
- Smartphones/tablets made cloud services portable.

Ex: Mobile banking apps, Google Drive.

Impact

- Flexibility and scalability for businesses.
- Enabled work-from-anywhere culture.
- Hybrid IT environments combine on-premise and cloud services.

Technology Drivers of Infrastructure Evolution



The evolution of IT infrastructure is driven by several key technological advancements that enable faster more efficient & interconnected systems

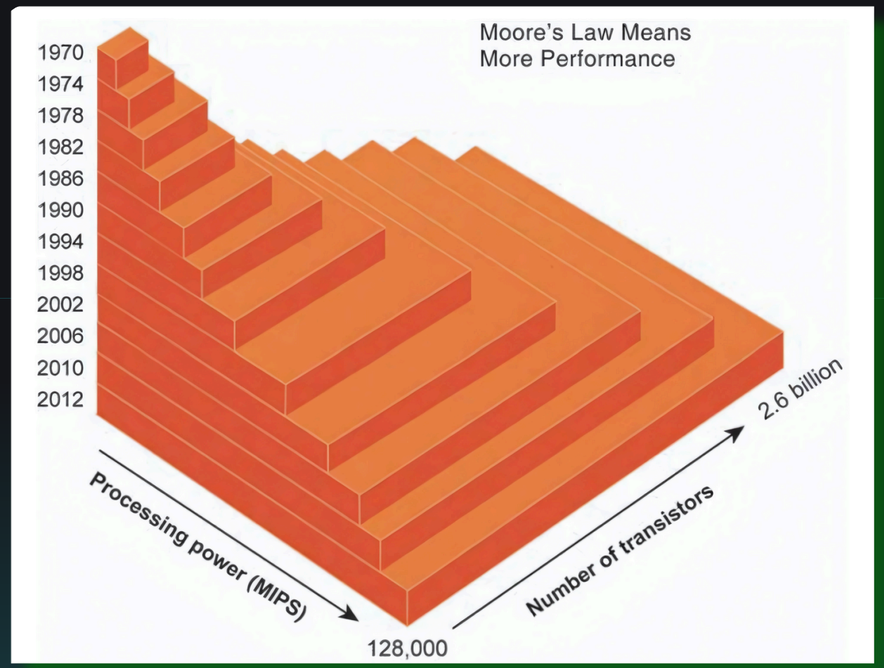
Moore’s Law

What it is

Computer chips become faster, more powerful, and cheaper every few years.

Why it Matters

This lets us run advanced software and process data quickly on smaller, cheaper devices.



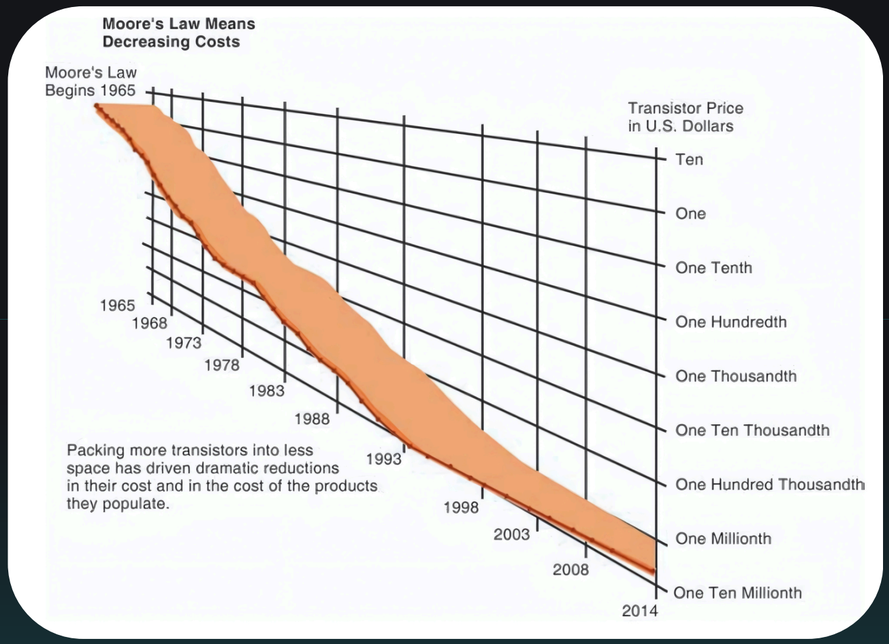
Mass Digital Storage

What it is

The cost of storing digital data is decreasing, while storage capacity is increasing.

Why it Matters

We can store large amounts of data cheaply, making it easier to save videos, run cloud services, and analyze big data.



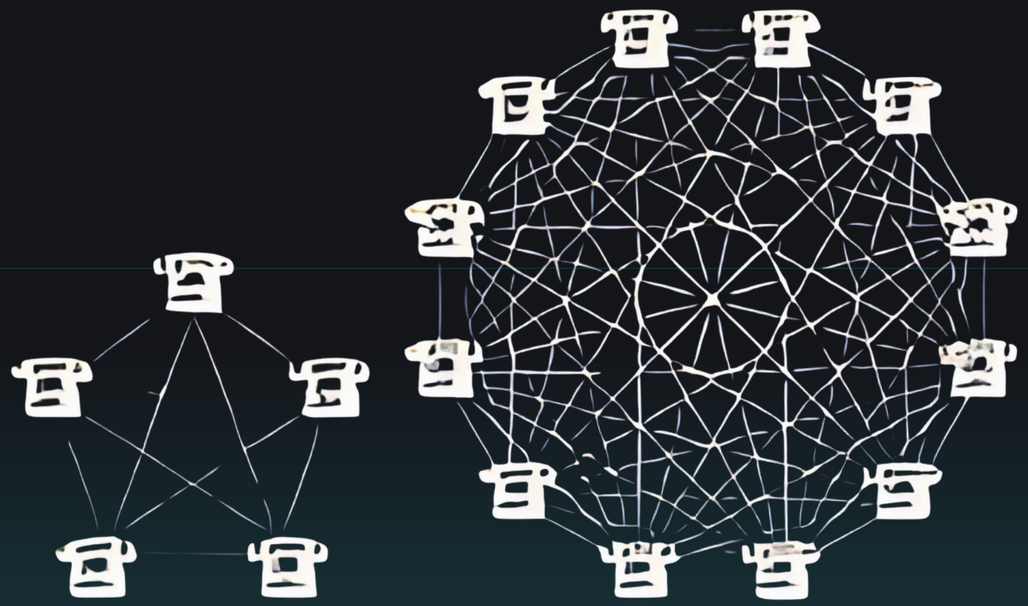
Metcalfe’s Law

What it is

A network (like the internet) becomes more valuable as more people or devices use it.

Why it Matters

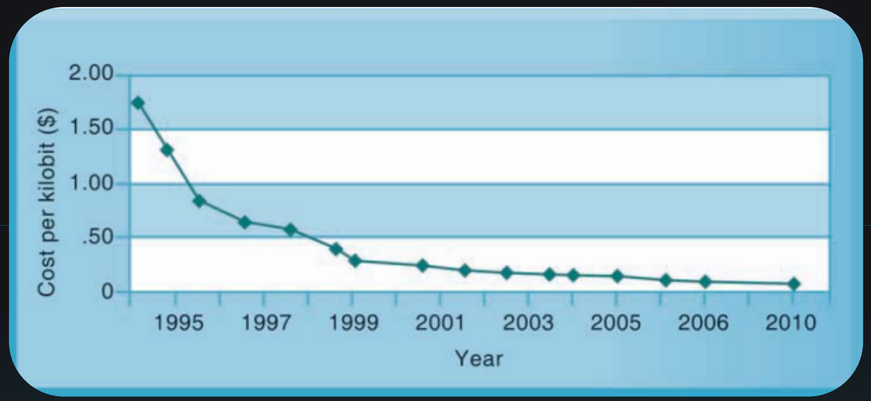
More users make networks like social media, online shopping, and smart devices more useful, attracting even more people.



Declining Communication Costs and Internet Growth

What it is

Sending data worldwide is cheaper and easier, and the internet is everywhere now.



Why it Matters

This connects people and businesses worldwide, supports remote work, boosts online shopping, and enables video calls and streaming.

Standards & Network Effects

What it is

Common rules like Wi-Fi and USB help technology work together, and the more people use something, the better it becomes.



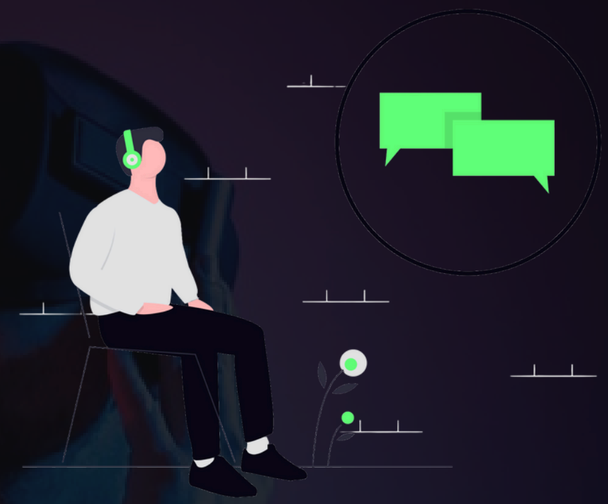
Why it matters

This makes devices and services work smoothly, supports innovation, and helps things like app stores and smart devices grow.

What are the IT Infrastructure Components



What are the IT Infrastructure Components?



Seven key parts must work together to create a strong IT system.
Below are the main technologies and their providers for each part.

- Computer Hardware Platforms
- Operating System Platforms
- Enterprise Software Applications
- Data Management & Storage





- **Networking / Telecommunication Platforms**
- **Internat Platforms**
- **Consulting System Integration services**





Computer Hardware Platforms

Client machines



- Desktop PCs, laptops

- Mobile computing: smartphones, tablets

Servers



- Blade servers: ultrathin computers stored in racks

Mainframes



-IBM mainframe equivalent to thousands of blade servers

Operating System Platforms

1 **Server level**

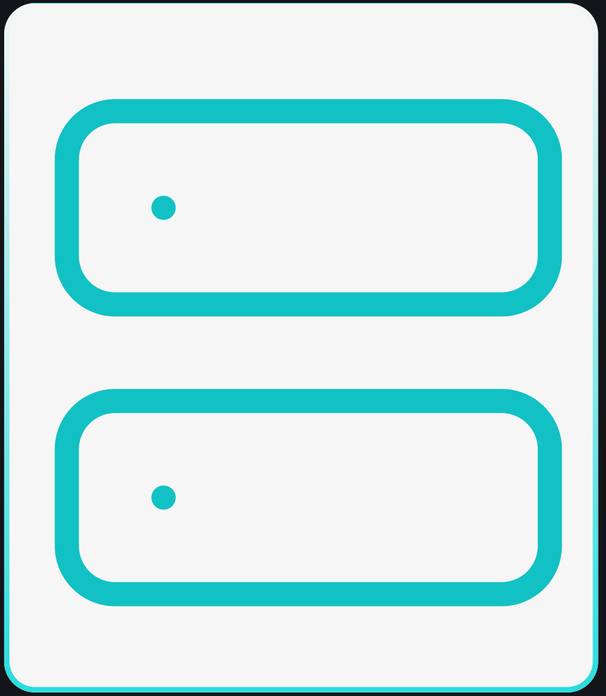
65% run Unix or Linux; 35% run Windows

2 **Client level**

— 90% run Microsoft Windows (Windows 8, Windows 7, etc.)

— Mobile/multitouch (Android, iOS)

— Cloud computing (Google's Chrome OS)



Enterprise Software Applications

- Enterprise Application Providers





- Middleware Providers





Data Management and Storage



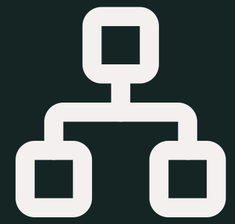
Database Software

- **IBM(DB2)**
- **ORACLE**
- **MICROSOFT (SQL Server)**
- **MYSQL**



Physical Data Storage

- EMC CORP (large-scale systems)
- SEAGATE
- WESTERN DIGITAL



Storage Area Networks

- Connect multiple storage devices on dedicated network

Networking/ Telecommunications Platforms



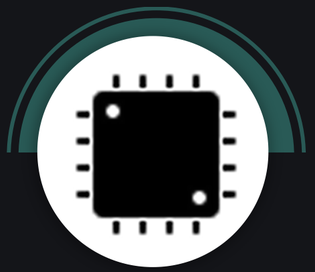
Telecommunication Services

Telecommunications
cable
AT&T
Verizon



Network Operating Systems

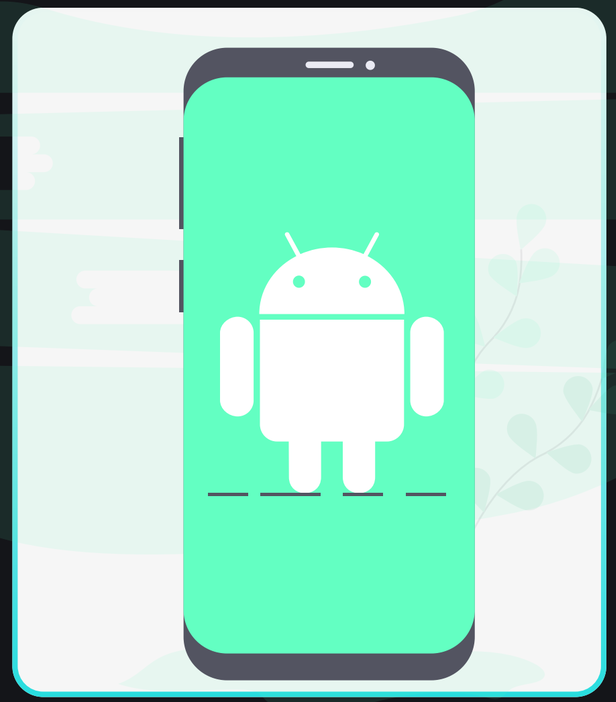
Windows Server
Linux
Unix



Network Hardware Providers

Cisco,
Alcatel-Lucent
Juniper Networks

Internet Platforms



1. Hardware, software, management services to support company Web sites (including Web-hosting services), intranets, extranets

2. **Internet hardware server market**

   — IBM, Dell, Sun(Oracle), HP

3. **Web development tools/suites**

   — Microsoft (Visual Studio and .NET), Oracle-Sun (Java), Adobe, Real Networks

Consulting and System Integration Services

Even large firms do not have resources for full range of support for new, complex infrastructure

- **Leading Consulting Firms** — Accenture, IBM Global Services, HP, Infosys, Wipro Technologies
- **Software Integration** — ensuring new infrastructure works with legacy systems
- **Legacy Systems** — Older TPS created for mainframes that would be too costly to replace or redesign

What are the Trends in Computer Hardware Platforms

Contemporary Hardware Platform Trends



1. The rise of advanced computer hardware and networking has transformed business operations.

2. **Mobile devices are now key players, taking on many roles traditionally performed by PCs text**

The Mobile Digital Platform

Mobile Devices



- Smartphones: iPhone, Android, and BlackBerry models.

- Capabilities: Data transmission, web surfing, email, instant messaging, and content sharing.

Netbooks & Tablets



- Lightweight netbooks optimized for wireless communication.

- Tablets (e.g., iPad) and e-book readers (e.g., Kindle).

E-book Readers



-Kindle, Kobo, Nook Capabilities: Store thousands of books in one device, Adjustable font size, brightness, and themes

Consumerization

Process by which new information technologies that are first adopted by consumers make their way into businesses



- Smartphones
- Tablets
- Applications like Gmail, Dropbox, and Google Apps

which were originally consumer-focused but are now widely used in corporate settings.

Bring Your Own Device (BYOD)

A specific aspect of IT consumerization where employees use their personal devices for work purposes, including accessing corporate networks and data.

PROS

- Cost savings for employers
- productivity and satisfaction
- Fosters Work-Life integration

CONS

- Security risks
- Policy and Compliance issues
- IT management challenges

Cloud Computing

Cloud computing delivers IT resources and services like storage, databases, and software over the internet

Types

IaaS ---------> Infrastructure as a Service (e.g., AWS EC2)

PaaS ---------> Platform as a Service (e.g., Google App Engine)

SaaS ---------> Software as a Service (e.g., Google Workspace)

Benefits

- Access resources anytime, anywhere.
- Reduces IT infrastructure costs.
- Supports innovation through scalable resources
- Faster deployment of applications and services.

Green Computing

Involves designing, using, and disposing of IT resources in an environmentally friendly manner.

Goals

- Reduce energy consumption.
- Minimize carbon footprint.
- Decrease electronic waste.

Importance

- Promotes environmental sustainability
- Saves operational costs for businesses.
- Ensures compliance with environmental regulations (e.g., WEEE Directive)



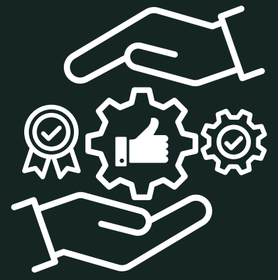
Environmental Impact

- High energy use from data centers and devices.
- Electronic waste as a growing global issue.



Innovative Trends

- Advanced cooling systems (e.g., liquid cooling).
- Edge computing for efficient resource use.



Best Practices

- Green IT procurement for energy-efficient devices.
- Educate employees on power-saving habits.

Power-Saving in Computer Hardware



Energy-Efficient Hardware

Use multicore processors and energy-efficient components.

Enable automatic sleep mode and power-down settings.

Power Management Features

Server Virtualization

Run multiple virtual servers on fewer physical machines to optimize energy use.

Leverage shared cloud resources for better energy efficiency.

Cloud Computing

Dynamic Voltage and Frequency Scaling (DVFS)

Adjust processor voltage and frequency to save energy.



What are Trends in Computer Software Platforms

Themes of Contemporary Software Platform Evolution

1. Linux and open-source software
2. Java, HTML, and HTML5
3. Web services and service-oriented architecturee
4. Software outsourcing and cloud services

What is Open-source Software?

Software anyone can use, modify, and share freely.
Also, it is not restricted to any specific operating system or hardware technology.
Most open-source software is based on a Linux or Unix operating system.









LINUX

Linux, created by Finish programmer Linus Torvalds in 1991, an operating system related to Unix.

This works on all the major hardware platforms from mainframes to servers to clients.



Why Businesses Choose Linux

- Cost-effective solutions
- Secure and stable
- Highly customizable
- Adopted by major companies

Enterprise Applications

- **Java Applets**
  Small programs embedded in web pages, running in browsers

- **Web Browsers**
  Tools with graphical interfaces to display web pages & access online resources

- **Enterprise Applications**
  Supports complex e-commerce and e-business applications
  facilitates communication with back-end transaction systems

HTML



HTML, Web Page Description Language for structuring web content.

This enables seamless navigation through dynamic links.

HTML5 Evolution

- Embeds multimedia (audio, video) without add-ons
- Enhances cross-device compatibility and offline data storage

HTML5 Benefits

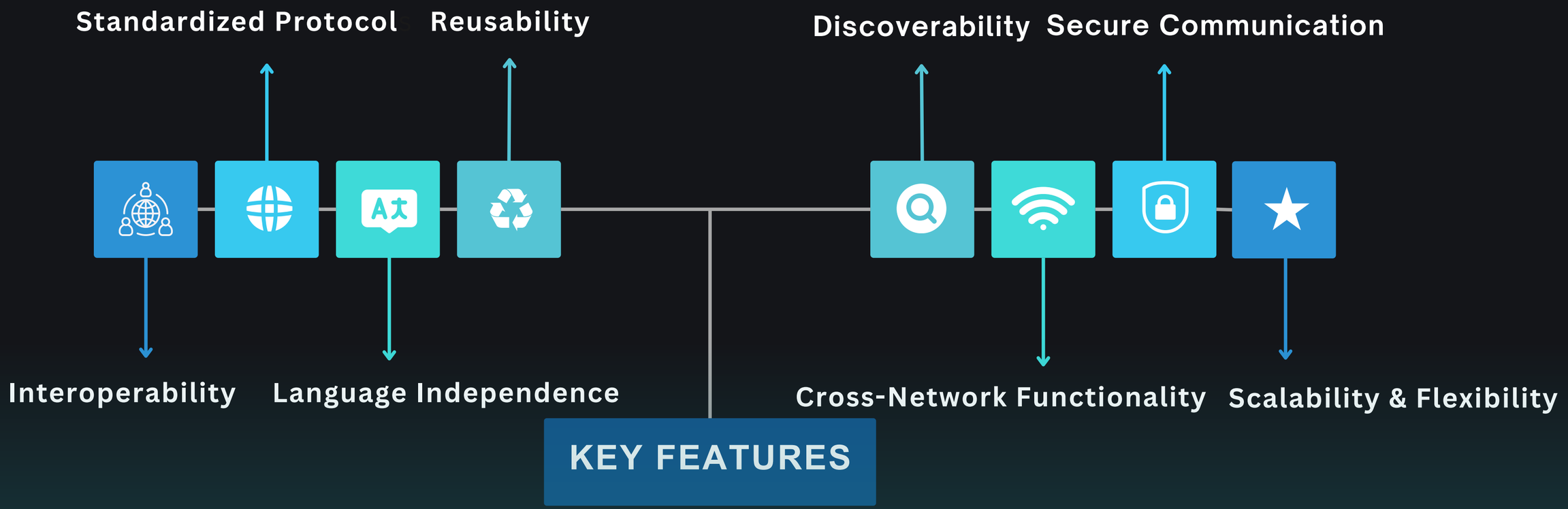
- Faster execution with app-like features
- Supported by major tools (Safari, Chrome, Gmail) and popular websites





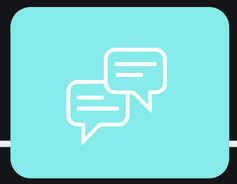
Web Services

Set of loosely coupled software components that exchange information with each other using universal Web communication standards and languages.



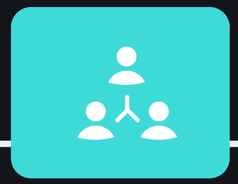
Extensible Markup Language

A flexible text-based format designed to store, transmit, and structure data in a way that is both human-readable and machine-readable.



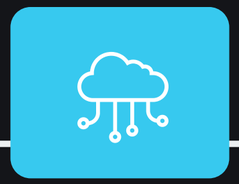
MEDIUM FOR COMMUNICATION

Communicates using XML messages over standard web protocols (HTTP/HTTPS)



SMOOTH CONNECTION

Connection between applications without custom programming



OPEN-STANDARD SYSTEMS

Enables compatibility and interoperability across platforms



LINK ENTERPRISE SYSTEMS

Integrate systems for ERP, order processing, and key business operations

Service Oriented Architecture

SOA is a set of self-contained services that communicate to create working software applications.

KEY BENEFITS

- Reusability
- Compatibility
- Simplicity
- Cost-Effectiveness
- Improved Reliability

IBM WebSphere

Microsoft® .NET

Software Outsourcing & Cloud Services

Many businesses continue using legacy systems because they are cost-effective. However, Businesses now prefer to rent or outsource software for flexibility, scalability, and cost savings.

External Sources for Software:

01



Software Packages & Enterprise Software

02



Software Outsourcing

03



Enhanced User Satisfaction

Software Packages & Enterprise Software



Software packages are ready-made programs businesses use for tasks like paying employees or managing orders. They are key to modern systems

SAP and Oracle-PeopleSoft offer software for tasks like

- warehousing , CRM
- supply chain , finance
- HR

These are more affordable than creating custom solutions.

Software Outsourcing

Hiring an external company or team to handle software development tasks or projects instead of managing them in-house.

Types of Services in Software Outsourcing

- Custom Software Development
- Mobile App Development
- Web Development
- IT Consulting
- QA & Testing
- Maintenance and Support

Reasons for Outsourcing

- Cost Efficiency: Reducing costs by outsourcing to regions with lower labor rates.
- Access to Expertise: Gaining access to skilled developers and technologies.Faster decision-making with shared data access.
- Scalability: Quickly scaling teams up or down based on project requirements.

Challenges of Outsourcing

- Communication Barriers: Differences in time zones, languages, or cultures.
- Quality Control: Ensuring the outsourced team meets quality expectations.
- Data Security: Protecting intellectual property and sensitive data.
- Management Issues: Aligning outsourced teams with company objectives and processes.

Cloud-Based Software Services & Tools

What is Cloud Based Software?

Hosted on powerful servers in data centers & accessible via the internet using a web browser

Shift from Traditional Software

Past ------> Locally installed (Microsoft Word, Adobe Illustrator)

Present ------> Cloud services for individuals and businesses

Software as a Service (SaaS)

Salesforce.com for customer relationship management

Service Level Agreement (SLA)

Provider’s responsibilities & service expectations

Mashups & Apps

What is Mashup



- Combines multiple online services or data sources
- Creates a new application or tool that adds more value

Eg: ZipRealty
Combines Google Maps and Zillow real estate data

What is App

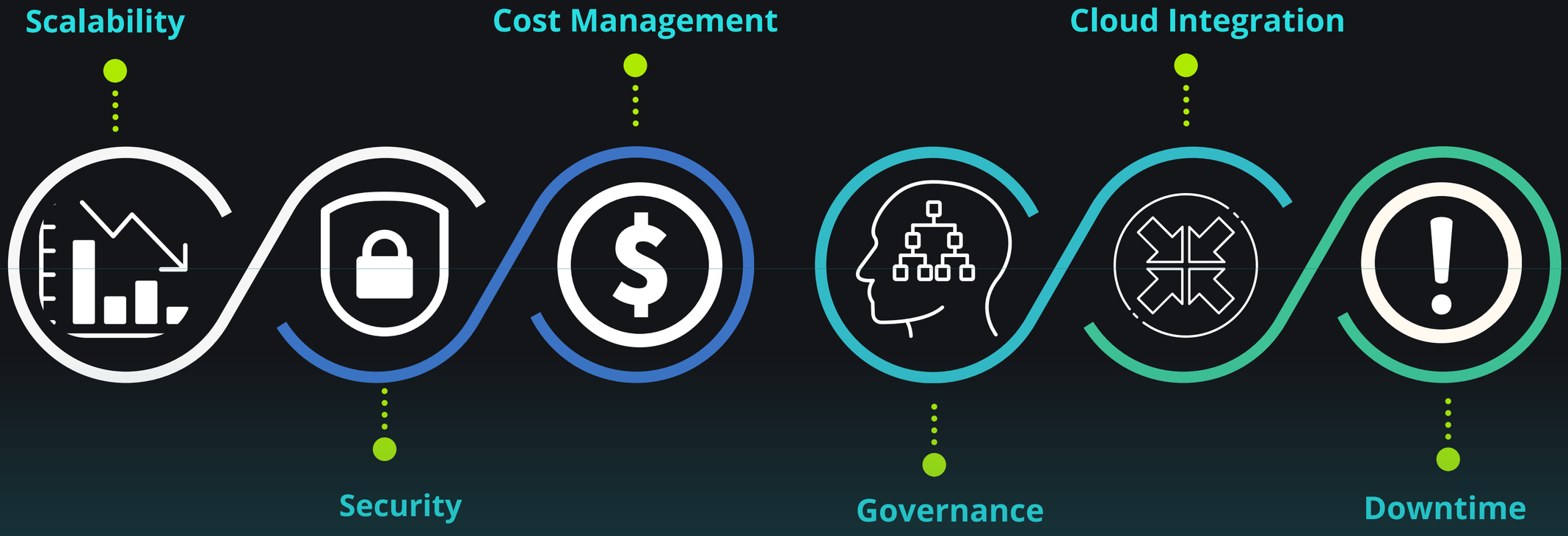


- Small software tools running on smartphones, tablets, or computers
- Tasks: Entertainment, productivity, business, etc

Eg: Mobile apps for games, social media, business

What is Managing IT Infrastructure & Management Solutions

IT Infrastructure Management Challenges



Adapting to Change in IT Infrastructure

Scalability



The ability of systems to expand to serve growing business demands without failures.

Planning for mergers, business growth, or seasonal workload changes

Policy Strategy for Mobile & Cloud



Managing mobile devices and cloud computing through policies and procedures.

Inventory tracking, security updates, and data access rules

Contractual Arrangements



Ensuring cloud service agreements meet availability, security, and performance needs.

Clear SLAs for mission-critical applications

Strategic IT Investments

Total Cost of Ownership (TCO)

- Includes acquisition, maintenance, training, and operational costs.
- Example: TCO of mobile devices ranges from $1,000 to $3,000 annually.

Rent vs. Buy Decisions

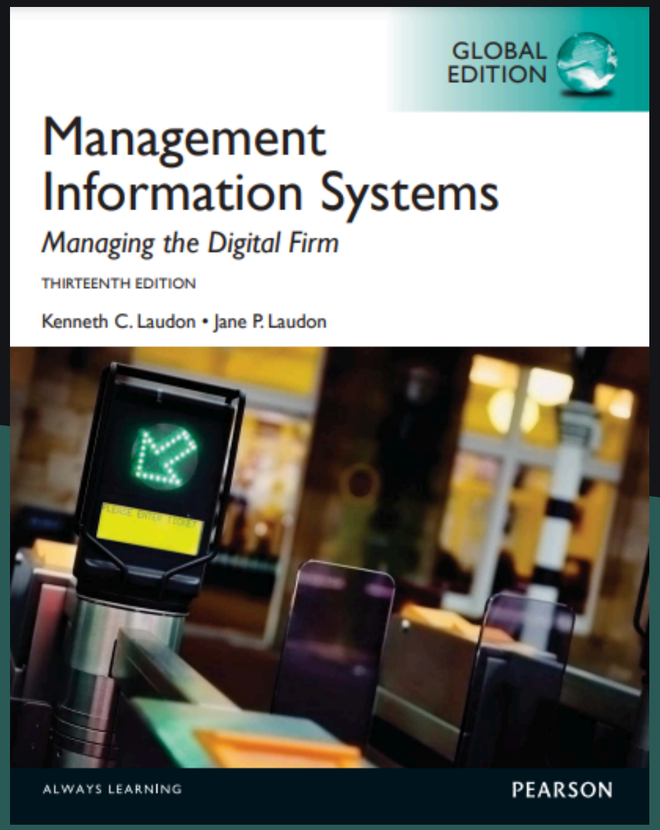
- Weighing the pros and cons of owning versus renting IT infrastructure.
- Example: Cloud services for scalability but higher long-term costs.



Competitive Benchmarking

- Assessing IT investment levels compared to competitors.
- Example: Identifying cost-effective practices without overspending.

References



Management Information System- **Kenneth C.Laudon , Jane P.Laudon**

Group Members

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THANK YOU !

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Q & A